

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 7
 - More on Closed Sets

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Adherent and Accumulation Points

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- suprema and infima are accumulation points!
- 0 is an accumulation point of $\{1/1, 1/2, 1/3, \dots\}$

Characterizing accumulation points

Theorem (Apostol Theorem 3.17)

A point $\vec{x}$ is an accumulation point of A if and only if for all $r > 0$, the ball $B(\vec{x}; r)$ contains infinitely many points of A .

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Obviously, if for all $r > 0$, the ball $B(\vec{x}; r)$ contains infinitely many points of A , then it contains at least one point of A different from $\vec{x}$, so it's an accumulation point.

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Suppose instead that $\vec{x}$ is an accumulation point and let $r > 0$. It suffices to show that the set $C = (B(\vec{x}; r) \setminus \{\vec{x}\}) \cap A$ is infinite.



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Suppose that C is finite.

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Suppose that C is finite.

Then the set $\{|\vec{y} - \vec{x}| : \vec{y} \in C\}$ is also finite.

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However, $\vec{y} \in C$ and $|\vec{y} - \vec{x}| < s$, contradicting the minimality of s .

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However, $\vec{y} \in C$ and $|\vec{y} - \vec{x}| < s$, contradicting the minimality of s .

We conclude that C is infinite. □

Closure of a set

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Theorem (Apostol Theorem 3.18, 3.20, 3.22)

A set is closed if and only if it contains all of its adherent points (ie. $A = \bar{A}$), or equivalently if and only if A contains all of its accumulation points.

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This implies that $B(\vec{x}; r) \subseteq \mathbb{R}^n \setminus A$.

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Since $\vec{x}$ was an arbitrary element of $\mathbb{R}^n \setminus A$, this proves $\mathbb{R}^n \setminus A$ is open.

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Therefore A is closed.



Closures are closed

Theorem

If $A \subseteq \mathbb{R}^n$ is any set, then $\overline{A}$ is closed.

Proof.

Exercise! □

Closed balls are closed

Problem

Show that the closed ball

$$\overline{B}(\vec{x}; r) = \{\vec{y} \in \mathbb{R}^n : |\vec{x} - \vec{y}| \leq r\}$$

is a closed set.